# Selena Wang

Updated: November 2, 2026
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*Network Science, Brain Imaging, Bayesian Analysis, Psychometrics.*

## Major Research Interests

### Methodology

- Neuroimaging, connectome-behavior linking
- Imaging genetics, imaging-genetics linking
- Social network analysis, network neuroscience
- High dimensional data analysis, clustering
- Psychometrics, item response theory, latent factor models
- Data integration
- Statistical machine learning, statistical philosophy

### Application

- Childhood psychopathology and disorders
- Neurodegeneration through aging and Alzheimer's disease
- Social factor of health outcomes

## Appointments

**Indiana University**
Assistant Professor, Department of Biostatistics and Health Data Science — Current
Director, School of Nursing Biostatistics Core
Faculty Member, Indiana Alzheimer's Disease Research Center
Faculty Member, Stark Neuroscience Institute
Faculty Affiliate, Center for Computational Biology and Bioinformatics
Faculty Affiliate, Center for Complex Networks and Systems Research

**Yale University**
Postdoctoral Associate, Department of Biostatistics — 2024

## Education

**The Ohio State University**
Ph.D. in Quantitative Psychology, Department of Psychology — 2022
- Thesis: "Modeling dependence between a network and item responses."

M.S. in Statistics
- Thesis: "Joint analysis of social and item response networks with latent space models."

M.A. in Quantitative Psychology
- Thesis: "When reliability does not signal reliable detection of experimental effects."

**University of Minnesota, Twin Cities** — 2017
B.S. in Psychology, Summa cum Laude with Highest Distinction

## Peer-reviewed Publications

*indicates equal contribution. †indicates publication as a mentor/senior author. indicates student/mentee under mentorship.

### Methodology/Application

1. **Wang, S.**, Zhang, X, Liu, Y., Xu, W., Tian, X., and Zhao, Y. (2026b). Latent space-based network analysis for brain–behavior linking in neuroimaging. *Nature Methods*, 23(1):225–235 (arXiv, paper)

2. **Wang, S.**, Sweet, T. M., and Paul, S. (2026a). The co-varying ties between networks and item responses via latent variables. *Psychometrika*, 91(2):641–668 (paper)

3. Xu, W., **Wang, S.**, Gao, S., Tian, X., Tan, C., Shen, X., Luo, W., Constable, T., Li, T., and Zhao, Y. (2025). Supervised brain node and network construction under voxel-level functional imaging. *Imaging Neuroscience*, 3:IMAG.a.56 (arXiv, paper)

4. Powla, P., Jayaprakash, T, Cornelison, C., Sweet, T., Shen, L., Gao, S., Yu, M., Wu, Y.-C., Zhao, Y., Saykin, A. J., and † **Wang, S.** (2025). Sex differences in the diffusion of tau neurofibrillary tangles across functional brain networks (abstract). *Alzheimer's & Dementia*, 21:e100019 (abstract)

5. Jayaprakash, T., Cornelison, C., Powla, P., Zhang, L., Fredericks, C. A., Saykin, A. J., and † **Wang, S.** (2025). Deep learning framework for characterizing tau-pet spatial heterogeneity along ad spectrum: A self-supervised approach (abstract). *Alzheimer's & Dementia*, 21:e109949 (abstract)

6. Cornelison, C., Wen, Q., Powla, P., Jayaprakash, T., Risacher, S., Saykin, A. J., Wu, Y.-C., and † **Wang, S.** (2025). Sex differences in structural connectivity-based ad pathology in the indiana memory and aging cohort (abstract). *Alzheimer's & Dementia*, 21:e103077 (abstract)

7. Zhang, X., Hulvershorn, L. A., Constable, T., Zhao, Y., and † **Wang, S.** (2025). Cost efficiency of fmri studies using resting-state vs. task-based functional connectivity. *Human Brain Mapping*, 46(9):e70260 (arXiv, paper)

8. Sweet, T.* and **Wang, S.*** (2025). Network science in psychology. *Psychological Methods* (paper)

9. **Wang, S.**, Wang, Y., Xu, F. H., Shen, L., and Zhao, Y. (2024a). Sex-specific topological structure associated with dementia identified via latent space network analysis. *Alzheimer's & Dementia*, 20(S2):e090404 (abstract)

10. **Wang, S.**, Wang, Y., Xu, F. H., Shen, L., Zhao, Y., and Alzheimer's Disease Neuroimaging Initiative (2025). Establishing group-level brain structural connectivity incorporating anatomical knowledge under latent space modeling. *Medical Image Analysis*, 99:103309 (paper, code)

11. **Wang, S.**, Wang, Y., Xu, F. H., Tian, X., Fredericks, C. A., Shen, L., Zhao, Y., and Alzheimer's Disease Neuroimaging Initiative (2024b). Sex-specific topological structure associated with dementia via latent space estimation. *Alzheimer's & Dementia*, 20(12):8387–8401 (paper)

12. Xu, W., **Wang, S.**, Shen, L., and Zhao, Y. (2024). Collaborative survival analysis on predicting alzheimer's disease progression. *Statistics in Biosciences*, 18(3):797–820 (paper)

13. Qiu, W., Chu, H., **Wang, S.**, Zuo, H., Li, X., Zhao, Y., and Ying, Z. (2024). Learning high-order relationships of brain regions. *International Conference on Machine Learning 2024 (27.5% acceptance rate)* (conference paper)

14. Tian, X., Wang, Y., **Wang, S.**, Zhao, Y., and Zhao, Y. (2024). Bayesian mixed model inference for genetic association under related samples with brain network phenotype. *Biostatistics*, 25(4):1195–1209 (paper)

15. **Wang, S.**, Paul, S., and De Boeck, P. (2023b). Joint latent space model for social networks with multivariate attributes. *Psychometrika*, 88(4):1197–1227 (paper, code)

16. **Wang, S.**, De Boeck, P., and Yotebieng, M. (2023a). Heywood cases in unidimensional factor models and item response models for binary data. *Applied Psychological Measurement*, 47(2):141–154 (paper, arXiv)

17. **Wang, S** and Edgerton, J. (2022). Resilience to stress in bipartite networks: application to the islamic state recruitment network. *Journal of Complex Networks*, 10(4):cnac017 (paper)

18. **Wang, S** and De Boeck, P. (2022). Understanding the role of subpopulations and reliability in between-group studies. *Behavior Research Methods*, pages 1–16 (paper)

19. **Wang, S.**, Wu, H., and Pek, J. (2022). Performance of alternative regression weights in the context of prediction versus inference (abstract). *Multivariate Behavioral Research*, 57(1):163–163 (abstract)

20. **Wang, S.** (2021). Recent integrations of latent variable network modeling with psychometric models. *Frontiers in Psychology*, 12 (paper)

### Collaborations

21. Smith, A. B., Knopf, A., Zhu, Y., Webster-Dekker, K. E., Moore, S. E., **Wang, S.**, and Wierenga, K. L. (2026). Healthcare-related social determinants of health are associated with pain severity among adults with comorbid heart failure and chronic pain. *BMC Health Services Research*, 26(1):1387 (paper)

22. Vilor-Tejedor, N., Danso, S., Albanus, R. D., Billingsley, K., Evans, T. E., Lee, L. Y., **Wang, S.**, Jiang, J., Liu, H., Ross, J., et al. (2026). Advancing global dementia research through equity and inclusion. *Alzheimer's & Dementia*, 22(1):e71069 (paper)

23. Smith, A., Knopf, A., Webster-Dekker, K., Moore, S., **Wang, S.**, and Wierenga, K. (2025). Healthcare-related social determinants of health are associated with pain severity among adults with comorbid heart failure and chronic pain (abstract). *Circulation*, 152(Suppl_3):A4344584–A4344584 (abstract)

24. Abughofah, Y., Risacher, S. L., Unverzagt, F. W., Farlow, M. R., Apostolova, L. G., Brosch, J. R., Clark, D. G., Mathew, S., Gao, S., **Wang, S.**, et al. (2025). The cognitive change index in the alzheimer's disease research center setting: Self-and informant-ratings for perceived cognitive decline. *Alzheimer's & Dementia*, 21(10):e70754 (paper)

25. Misra, S., **Wang, S.**, Quinn, T. J., Dawson, J., Zelano, J., Tanaka, T., Grotta, J. C., Khan, E., Beriwal, N., Funaro, M. C., et al. (2025). Antiseizure medications in poststroke seizures: A systematic review and network meta-analysis. *Neurology*, 104(3):e210231 (paper)

26. Miller, A. A., Sharp, E. S., **Wang, S.**, Zhao, Y., Mecca, A. P., van Dyck, C. H., and O'Dell, R. S. (2024). Self-reported hearing loss is associated with faster cognitive and functional decline but not diagnostic conversion in the adni cohort. *Alzheimer's & Dementia*, 20(11):7847–7858 (paper)

27. Li, J. S., Tun, S. M., Ficek-Tani, B., Xu, W., **Wang, S.**, Horien, C. L., Toyonaga, T., Nuli, S. S., Zeiss, C. J., Powers, A. R., Zhao, Y., Mormino, E. C., and Fredericks, C. A. (2024). Medial amygdalar tau is associated with mood symptoms in preclinical alzheimer's disease. *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging*, 9(12):1301–1311 (paper)

28. Misra, S., Dawson, J., Kwan, P., Kasner, S. E., Grotta, J. C., Larsson, D., Tanaka, T., Zaveri, H., **Wang, S.**, Khan, E. I., et al. (2024). Antiseizure medications in post-stroke seizures: A systematic review and network meta-analysis (p8-1.012). In *Neurology*, volume 102, page 5192. AAN Enterprises (paper)

29. Mishra, N. K., Kwan, P., Tanaka, T., Sunnerhagen, K. S., Dawson, J., Zhao, Y., Misra, S., **Wang, Selena**, Sharma, V. K., Mazumder, R., et al. (2023). Clinical characteristics and outcomes of patients with post-stroke epilepsy: protocol for an individual patient data meta-analysis from the international post-stroke epilepsy research repository (ipserr). *BMJ open*, 13(11):e078684 (paper)

30. Jamrozik, A., Clements, N., Hasan, S. S., Zhao, J., Zhang, R., Campanella, C., Loftness, V., Porter, P., Ly, S., **Wang, S.**, et al. (2019). Access to daylight and view in an office improves cognitive performance and satisfaction and reduces eyestrain: A controlled crossover study. *Building and Environment*, 165:106379 (paper)

## Grant Review

- Member, NIH Special Emphasis Panel on F (20) Fellowships: Alzheimer's and Neurodegeneration (NOV) 2026
- Member, NIH Special Emphasis Panel on F (20) Fellowships: Alzheimer's and Neurodegeneration (JUN) 2026
- Reviewer, GNPC Proteomics Data Challenge, Alzheimer's Disease Data Initiative 2026
- Member, NIH study section on Neuro Informatics, Computational and Data Analysis 2026
- Reviewer, NIH Autism Data Science Initiative 2025
- Reviewer, NSF Methodology, Measuremt & Stats Proposal 2024
- Reviewer, President's Research Excellence (PRE) program Catalyst concept papers, Ohio State 2022

## Grants

### Active

- **Deep Learning Embedding for Alzheimer's Disease**
  Funding Agency: Alzheimer's Disease Data Initiative
  Contribution: In-kind
  Role: Principal Investigator

- **Personalized Medicine under the Co-evolution of Connectomes and Pathological Proteins**
  Funding Agency: NIH
  Award Number: 1R01AG095242-01A1
  Period: 07/2026–06/2031
  Role: Principal Investigator
  Effort: 4.80 person-months/year (40%)

- **Short-term practice effects in Subjective Cognitive Decline: Moderators, prognostic value, and relationship to biomarker status**
  Funding Agency: NIH/NIA
  Award Number: 1R01AG099149-01
  Period: 03/2026–01/2031
  Role: Co-Investigator
  Effort: 0.60 person-months/year (5%)

- **Mechanics of Brain Folding in Human Infants Evaluated Using MRI**
  Funding Agency: NIH
  Award Number: 5R01NS133116-03
  Period: 08/2024–07/2029
  Role: Co-Investigator
  Effort: 0.60 person-months/year (5%)

- **School of Nursing Biostatistical Support**
  Funding Agency: Indiana University internal funding
  Period: 12/2024–06/2027
  Role: Co-Investigator
  Effort: 2.4 person-months in 2027 (20%)

- **Indiana Alzheimer's Disease Research Center**
  Funding Agency: NIH
  Period: 07/2026–06/2031
  Role: Co-Investigator
  Effort: 1.8 person-months/year (15%)

## Pending

- **Uncovering brain-wide molecular determinants of resilience across lifespan**
  Funding Agency: NIH Directors New Innovator Award Program
  Period: 09/2026–08/2031
  Role: Principal Investigator
  Effort: 4.8 person-months/year (40%)

- **Novel Computational Approach for Biological Mechanisms of ADRD Progression**
  Funding Agency: NIH/NIA
  Role: Principal Investigator

- **Individual differences in the biological mechanisms of ADRD progression explained by molecular and cellular vulnerabilities**
  Funding Agency: NIH
  Period: 12/2026–11/2031
  Role: Principal Investigator
  Effort: 4.80 person-months/year (40%)

- **Innovative Approaches to Measuring the Clinical Impact of Urinary Tract Infections in Children and Young Adults with Vesicoureteral Reflux**
  Funding Agency: NIH
  Period: 04/2027–03/2032
  Role: Co-Investigator
  Effort: 1.80 person-months/year (15%)

- **Predicting Youth Substance Use Onset and Escalation Through Neurobehavioral Mechanisms of Risky Decision-Making**
  Funding Agency: NIH
  Period: 04/2027–03/2032
  Role: Co-Investigator
  Effort: 1.20 person-months/year (10%)

- **Adipokine Signatures of Heart Failure: Integrating Leptin, Adiponectin, and LAR with Symptoms, Function, and Clinical Outcomes**
  Funding Agency: NIH
  Period: 12/2026–11/2031
  Role: Co-Investigator
  Effort: 1.20 person-months/year (10%)

- **Geriatrics Resources for Assessment and Care of Elders with HIV (GRACE-HIV): A Holistic Care Intervention to Improve Quality of Life for Older Adults with HIV**
  Funding Agency: NIH
  Period: 12/2026–11/2031
  Role: Co-Investigator
  Effort: 1.20 person-months/year (10%)

- **Reimagining the Approach to Vesicoureteral Reflux and Urinary Tract Infections in Children**
  Funding Agency: PCORI
  Period: 12/2026–11/2031
  Role: Co-Investigator
  Effort: 1.20 person-months/year (10%)
- **Blood-based biomarker utilization: Role of clinician characteristics and impact on clinical practices**
  Funding Agency: NIH/NIA
  Period: 10/2026–09/2027
  Role: Co-Investigator
  Effort: 0.60 person-months/year (5%)
- **PARTNER VET-AD: PARTNERing with VETerans to Accelerate the Early Detection of Alzheimer's Disease and Related Dementias**
  Funding Agency: Department of Defense
  Period: 10/2026–09/2030
  Role: Co-Investigator
  Effort: 0.60 person-months/year (5%)
- **Indiana AITC for Alzheimer's and Aging Disease Drug Discovery (I-A3D3)**
  Funding Agency: NIH
  Period: 07/2026–06/2031
  Role: Co-Investigator
  Effort: 0.60 person-months/year (5%)

## Not Funded

- **Rethinking network biology in the age of foundation models**
  Funding Agency: Beckman Young Investigator Program
  Role: Principal Investigator
- **Biological simulations of tau spreading on connectomes to explain an individual's resilience against developing Alzheimer's disease and related dementias**
  Funding Agency: Indiana Alzheimer's Disease Research Center
  Role: Principal Investigator
- **Novel AI Methods for Explaining Sex Disparities in Aging**
  Funding Agency: Alzheimer's Association Research Grant
  Role: Principal Investigator
- **Cost efficiency of fMRI studies in the diagnosis of ADHD using resting-state vs task-based functional connectomes**
  Funding Agency: Biostatistics, Epidemiology, and Research Design (BERD) Program Pilot Grant
  Role: Principal Investigator
- **Sex differences in Alzheimer's disease**
  Funding Agency: Gates Foundation
  Role: Principal Investigator
- **Novel computational models to identify protective mechanisms of ADRD**
  Funding Agency: Department of Defense
  Role: Principal Investigator
- **Precision Medicine under Functional and Structural Imaging Trajectory**
  Funding Agency: Ralph W. and Grace M. Showalter Research Trust Fund
  Role: Principal Investigator
- **Generative embedding of multi-scale neuroimaging for Alzheimer's disease**
  Funding Agency: 2025 ORAU Ralph E. Powe Junior Faculty Enhancement Awards Program
  Role: Principal Investigator

- **Mechanisms of neuronal changes under the influence of pathological proteins**
  Funding Agency: Klingenstein Fellowship Awards in Neuroscience
  Role: Principal Investigator

- **Neural Mechanism of Cognitive Resilience under the Influence of Tau**
  Funding Agency: Whitehall Foundation
  Role: Principal Investigator

- **Trajectory-based Multimodality Brain-age Delta AD Prediction across Life Span**
  Funding Agency: National Institute on Aging
  Application: 2023 NIH/NIA K01; Impact Score: 48
  Role: Principal Investigator

- **Phenotypes and Inflammatory Biomarkers of Secondary Musculoskeletal Pain in Heart Failure**
  Funding Agency: NIH
  Role: Co-Investigator

- **A Multi-Dimensional Toolkit to Assess Treatment Outcomes for Menopause Palpitations: A Population Based Study Integrating Physiological and Patient Reported Data**
  Funding Agency: Department of Defense / USAMRAA
  Role: Co-Investigator

- **Indiana Center for Autism Research**
  Funding Agency: Health Resources and Services Administration
  Role: Co-Investigator

- **Role of immune cell exhaustion in dementia risk post-traumatic brain injury**
  Funding Agency: Department of Defense
  Role: Co-Investigator

- **Vaginal Microbiome, Inflammatory Mediators, Metabolome, and Dysmenorrhea Symptom-based Phenotypes**
  Funding Agency: Department of Defense
  Role: Co-Investigator

## Teaching

- **Lead instructor of the course** at Indiana University — Fall 2026
  *SEMINAR: BIOSTATISTICS (DENT G910)*
  - Developed syllabus and materials
  - Oversaw student learning of statistical concepts
  - Provided hands on supporting with coding
- **Guest instructor of the course** at Indiana University — Fall 2025, 2026
  *Introduction to Quantitative Biomedical Sciences (Q581)*
  - Developed course materials
  - Delivered lecture
- **Lead instructor of the course** at Indiana University — Fall 2025
  *Biostatistics for Public Health I (PBHLP 551)*
  - Developed syllabus and materials
  - Oversaw student learning of statistical concepts
  - Provided hands on supporting with coding
- **Lead instructor of the course** at the Ohio State University (SEI available) — Fall 2020, Spring 2021
  *Introduction to Psychology (PSYCH 1100)*

  - Lead online courses during pandemic
  - Develop syllabus
  - Design course assignments
  - Deliver live lectures
- **Teaching Assistant** at the Ohio State University Fall 2018 –Spring 2020
  *Quantitative and Statistical Methods in Psychology (PSYCH 3321)*
  *Data Analysis in Psychology (PSYCH 2220)*
  - Design and grade course assignments
  - Quest lecture at weekly lab and review sessions
  - Tutor students and hold office hours

## Softwares

- **JNIRM:** An R package estimating the joint network and item response model.
  Link: https://github.com/selenashuowang/JNIRM.git
- **jlsm:** An R package estimating joint latent space models for social networks and multivariate attributes using a fast inference approach.
  Link: https://github.com/selenashuowang/jlsm.git
- **ABCModel:** An R package estimating the attributes-informed brain connectivity model.
  Link: https://github.com/selenashuowang/ABCModel.git
- **LatentSNA:** An R package estimating the latent variables-assisted statistical network analysis model.
  Link: https://github.com/selenashuowang/LatentSNA.git

## Honours, Awards and fellowships

- Research Career Institute in the Mental Health of Aging (CIMA) scholar 2026
- William H. Gates Sr. Fellow, Alzheimer's Disease Data Initiative 2025-2027
- REC Scholar, Indiana Alzheimer's Disease Research Center 2025
- DETxDigital Harbor Fellowship in Computational Psychometrics, Duolingo English Test 2021
- Finalist, Student Paper Competition, IISA 2021
  - Project: Joint latent space model for social networks with high-dimensional multivariate attributes.
- Ray Travel Award (for Service and Scholarship), Graduate Student Council, Ohio State 2020
- Graduate Student Conference Presentation Award, Ohio State 2018, 2019
- The International Meeting of the Psychometric Society Travel Award 2019
  - Project: The integration of social network analysis with psychometric models.
- Algorithms for Threat Detection (ATD) Travel Award 2018
  - Project: Joint Analysis of Social and Item Response Networks with Latent Space Models.
- College of Arts and Sciences Division of Social and Behavioral Sciences (SBS) Fellowship, Ohio State 2018
- Distinguished University Fellowship, Ohio State 2017, 2022
- Donald G. Paterson Award, Minnesota Psychological Association 2017
- Innovative Community Building Grants Award, University of Minnesota 2016
- Undergraduate Research Opportunities Program Grant, University of Minnesota 2016
  - Project: Gender and racial measurement bias in vocational interest assessment.
- Global Excellence Scholarship, University of Minnesota (4-years of in-state tuition) 2013

# Presentations

*indicates equal contribution. †indicates publication as a mentor/senior author. indicates student/mentee presenter.

## Invited Presentations

1. L. Altan and †**S. Wang**, "Modulating Impact of Functional Connectomes on AD Pathology", *20th International Joint Conference on Computational and Financial Econometrics (CFE) and Computational and Methodological Statistics (CMStatistics; CFE-CMStatistics 2026), December 2026 (scheduled), HTW Berlin, Germany.*

2. **S. Wang**, "Improve power of brain-behavior studies using LatentSNA", *32nd Annual Meeting of the Organization for Human Brain Mapping (OHBM), June 2026, Bordeaux, France.*

3. Krishnendu Chandra and †**S. Wang**, "Mediating Impact of Functional Connectomes on AD Pathology via the Functional Graphical Variational Autoencoder (FUNGVA) Model", Math Department, *Indiana University, Indianapolis, May, 2026.*

4. **S. Wang**, "Implementing Network Science in Generative Statistical Processes", *The American Statistical Association (ASA) Statistics and Data Science in Aging (SDSA) Interest Group, Virtual Seminar, February, 2026.*

5. **S. Wang**, "New developments in implementing network science in connectome research", *Indiana University, Center for Computational Biology and Bioinformatics Seminar, December, 2025.*

6. **S. Wang**, "Neuroimaging connectivity analysis needs network science for brain-behavior linking", *University of Louisville, Department of Bioinformatics and Biostatistics Seminar, October, 2025.*

7. **S. Wang**, "New developments in implementing network science in connectome research", *19th International Joint Conference on Computational and Financial Econometrics (CFE) and Computational and Methodological Statistics (CMStatistics; CFE-CMStatistics 2025), December 2025, Birkbeck, University of London, UK.*

8. **S. Wang**, "Sex differences in functional connectomes associated with Alzheimer's disease (Poster)", *2025 Spring ADRC Meeting, May, 2025.*

9. **S. Wang**, "Joint network and psychometric models", *University of California, Los Angeles, Quantitative Psychology brown bag seminar, March, 2025.*

10. **S. Wang**, "Current projects in network-based connectome research", *Indiana Public Health Informatics, March, 2025.*

11. **S. Wang**, "Detect imaging biomarkers with statistical network analysis", *Center for Complex Networks and Systems Research, March, 2025.*

12. **S. Wang**, "Detect imaging biomarkers with statistical network analysis", *Adolescent Behavioral Health Research Program 2025, February, 2025.*

13. **S. Wang**, "Sex-specific topological structure associated with dementia via latent space estimation", *2025 ICSA Applied Statistics Symposium, June 2025.*

14. **S. Wang**, "Heritability and genetic contribution analysis of the co-development of FC and SC during childhood", *Eastern North American Region (ENAR) of the International Biometric Society 2025 Spring Meeting, March 2025, New Orleans, LA.*

15. **S. Wang**, "Linking social networks with behavior outcomes", *Indiana University Biostatistics and Health Data Science Departmental Seminar, November, 2024.*

16. **S. Wang**, Y. Wang, F. Xu, C. Fredericks, L. Shen and Y. Zhao, "Sex-specific topological structure associated with dementia via latent space estimation", *18th International Joint Conference on Computational and Financial Econometrics (CFE) and Computational and Methodological Statistics (CMStatistics; CFE-CMStatistics 2024), December 2024, King's College London, UK.*

17. **S. Wang**, "Neuroimaging connectivity analysis needs network science for brain-behavior linking", *Stark Neurosciences Research Institute, October, 2024.*

18. **S. Wang**, Y. Wang, F. Xu, C. Fredericks, L. Shen and Y. Zhao, "Sex-specific topological structure associated with dementia via latent space estimation", *2024 Statistical Methods in Imaging (SMI) Conference, May 2024, Indianapolis, IN.*

19. **S. Wang**, "Detect imaging biomarkers with statistical network analysis", *Statistics in Imaging Virtual Working Group webinar, November, 2024.*

20. **S. Wang**, "Detect imaging biomarkers with statistical network analysis.", *Purdue University CONNplexity Lab, August, 2024.*

21. **S. Wang**, "Detect imaging biomarkers with statistical network analysis", *Purdue and Indiana University Joint Seminar, October, 2024.*

22. **S. Wang**, "Inference-based statistical network analysis uncovers star-like brain functional architectures for internalizing psychopathology in children.", *Center for Computational Biology and Bioinformatics, October, 2024.*

23. **S. Wang**, "Detect imaging biomarkers with statistical network analysis", *University of Illinois Urbana-Champaign, Urbana-Champaign, January, 2024.*

24. **S. Wang**, "Detect imaging biomarkers with statistical network analysis", *Rush University, Chicago, January, 2024.*

25. **S. Wang**, "Detect imaging biomarkers with statistical network analysis", *Indiana University, Indianapolis, January, 2024.*

26. **S. Wang**, "Detect imaging biomarkers with statistical network analysis", *University of Minnesota, Online, December, 2023.*

27. **S. Wang**, "Detect imaging biomarkers with statistical network analysis", *University of Utah, Online, November, 2023.*

28. **S. Wang**, "Detect imaging biomarkers with statistical network analysis", *St Jude Children's Research Hospital, Online, November, 2023.*

29. **S. Wang**, "Detect imaging biomarkers with statistical network analysis", *Georgia Tech University, Atlanta, November, 2023.*

30. **S. Wang**, "The integration of psychometric and network models for behavior and neuroscience research", *University of Maryland, College Park, April, 2023.*

31. **S. Wang**, "The integration of psychometric and network models for educational research", *Michigan State University, February, 2023.*

32. **S. Wang**, "The integration of psychometric and network models for educational research", *University of Nebraska, Lincoln, January, 2023.*

33. **S. Wang**, J.R. Lockwood, and Y. Attali, "Fitting psychometric models to digital-first assessment", *Duolingo English Test, August, 2021.*

34. **S. Wang**, S. Paul, and P. De Boeck, "Joint latent space model for social networks with high-dimensional multivariate attributes.", *International Indian Statistical Association Conference, May, 2021.*

35. **S. Wang**, S. Paul, J. Logan, and P. De Boeck, "Joint modeling of network and psychometric data", *International Meeting of Psychometric Society, July, 2019.*

## National & Regional Conferences

36. A. Sharma, C. Cornelison, L. Altan, K. Chandra, Q. Wen, K. Nho, A. Saykin, Y. Wu and †**S. Wang**, "Proteomics signature of whole-brain microstructural integrity of white matter using LatentSNA (poster)", *Alzheimer's Association International Conference (AAIC) 2026, July 2026, London, UK, and online.*

37. C. Cornelison, A. Sharma, L. Altan, K. Nho, A. Saykin and †**S. Wang**, "Plasma proteomics biomarkers that differentiate asymptomatic from symptomatic Alzheimer's disease (poster)", *Alzheimer's Association International Conference (AAIC) 2026, July 2026, London, UK, and online. (accepted, not attended)*

38. G. Osei, S. Gao, A. Saykin and †**S. Wang**, "A Psychometric Evaluation of the Cognitive Change Index in the Indiana Memory and Aging Cohort (poster)", *Alzheimer's Association International Conference (AAIC) 2026, July 2026, London, UK, and online.*

39. H. Lee, T. Sweet and **S. Wang**, "A Multilevel Modeling Approach to Network Diffusion in Alzheimer's Disease (poster)", *Alzheimer's Association International Conference (AAIC) 2026, July 2026, London, UK, and online.*

40. H. Lee, T. Sweet and **S. Wang**, "A Multilevel Modeling Approach to Network Diffusion in Alzheimer's Disease", *International Meeting of the Psychometric Society (IMPS) 2026, July 2026, Seoul, Republic of Korea.*

41. C. Cornelison, A. Sharma, L. Altan, K. Nho, A. Saykin, and †**S. Wang**, "Plasma proteomics biomarkers that differentiate asymptomatic from symptomatic Alzheimer's disease", *11th Annual NHGRI Research Training and Career Development Meeting, March 2026, St. Louis, MO.*

42. L. Altan and †**S. Wang**, "Bayesian latent space co-evolution model to explain AD progression (poster)", *Eastern North American Region (ENAR) of the International Biometric Society 2026 Spring Meeting, March 2026, Indianapolis, IN.*

43. T. Jayaprakash and †**S. Wang**, "Alzheimer's Disease Subtype Identification from Spatial Tau-PET Heterogeneity Using Self-Supervised Deep Learning (poster)", *Eastern North American Region (ENAR) of the International Biometric Society 2026 Spring Meeting, March 2026, Indianapolis, IN.*

44. C. Cornelison, K. Chandra, T. Q. Wen, T. Jayaprakash, A. Saykin, Y. Wu, and †**S. Wang**, "Sex differences in structural connectivity-based AD pathology in the Indiana Memory and Aging cohort (poster)", *Eastern North American Region (ENAR) of the International Biometric Society 2026 Spring Meeting, March 2026, Indianapolis, IN.*

45. L. Altan, T. Jayaprakash, and †**S. Wang**, "Alzheimer's Disease Subtype Identification from Spatial Tau-PET Heterogeneity Using Self-Supervised Deep Learning", *Statistical Methods in Imaging (SMI) 2026 Conference, June 2026, University of Michigan, Ann Arbor, MI. (oral presentation)*

46. **S. Wang**, "Brain Resilience in Aging", *Research Career Institute in the Mental Health of Aging (CIMA) Conference, Scholar Presentation, March 2026.*

47. T. Jayaprakash, C. Cornelison, P. Powla, L. Zhang, C. Fredericks, A. Saykin, and †**S. Wang**, "Deep Learning Framework for Characterizing Tau-PET Heterogeneity in Alzheimer's Disease: A Self-supervised Approach (poster)", *Alzheimer's Association International Conference (AAIC) 2025, July 2025, Toronto, Canada, and online. (abstract)*

48. P. Powla, T. Jayaprakash, C. Cornelison, T. Sweet, L. Shen, S. Gao, M. Yu, Y.-C. Wu, Y. Zhao, A. J. Saykin and †**S. Wang**, "Sex Differences in the Diffusion of Tau Neurofibrillary Tangles Across Functional Brain Networks (poster)", *Alzheimer's Association International Conference (AAIC) 2025, July 2025, Toronto, Canada, and online. (accepted, not attended)*

49. **S. Wang**, "Foundational generative embedding models for neuromolecular networks in Alzheimer's Disease (poster)", *Alzheimer's Association International Conference (AAIC) 2025, July 2025, Toronto, Canada, and online.*

50. C. Cornelison, P. Powla, T. Jayaprakash, A. Saykin, Y. Wu, and †**S. Wang**, "Sex differences in structural connectivity-based AD pathology in the Indiana Memory and Aging cohort (poster)", *Alzheimer's Association International Conference (AAIC) 2025, July 2025, Toronto, Canada, and online. (abstract)*

51. **S. Wang**, Y. Wang, F. Xu, C. Fredericks, L. Shen and Y. Zhao, "Sex-specific topological structure associated with dementia identified via latent space network analysis (poster)", *Alzheimer's Association International Conference (AAIC) 2024, July–August 2024, Philadelphia, PA, and online. (abstract, poster)*

52. **S. Wang**, Y. Liu, W. Xu, X. Tian and Y. Zhao, "Inference-based statistical network analysis uncovers star-like brain functional architectures for internalizing psychopathology in children", *Eastern North American Region (ENAR) of the International Biometric Society 2024 Spring Meeting, March 2024, Baltimore, MD.*

53. **S. Wang**, Zhang, X., Y. Liu, W. Xu, X. Tian and Y. Zhao, "Unveiling star-Like brain networks in adolescent internalizing psychopathology: an inference-based statistical approach (Poster)", *Neuroscience Track Brunch and Poster Sessions, February, 2024.*

54. **S. Wang**, Y. Wang, F. Xu, L. Shen and Y. Zhao, "Establishing group-level brain structural connectivity incorporating anatomical knowledge under latent space modeling", *Statistical Methods in Imaging Conference 2023, May, 2023.*

55. W. Qiu, **S. Wang**, X. Li, Y. Zhao, R. Ying, "Understanding interactions of ROIs via learnable hyperedges (Poster)" , *Organization for Human Brain Mapping, July, 2023.*

56. **S. Wang**, Y. Wang, F. Xu, L. Shen and Y. Zhao, "Identifying gender based neuromarkers of Alzheimer's disease using ABC network model" (Poster), *Organization for Human Brain Mapping, July, 2023.*

57. P. De Boeck **S. Wang** and S. Cho, "Binary time series for eye fixations: the quality of parameter estimates", *National Council on Measurement in Education Conference, April, 2023.*

58. P. De Boeck and **S. Wang**, "Pervasive measurement bias", *Quantitative Methodology Center 2022 Conference, June, 2022.*

59. **S. Wang**, J.R. Lockwood, and Y. Attali, "Fitting psychometric models to digital-first assessment", *National Council on Measurement in Education, April, 2022.*

60. **S. Wang**, S. Paul, and P. De Boeck, "Modeling classroom outcome with network analysis", *National Council on Measurement in Education, April, 2022, cancelled.*

61. **S. Wang**, S. Paul, and P. De Boeck, "Joint latent space model for social networks with high-dimensional multivariate attributes", *International Meeting of Psychometric Society, July, 2021.*

62. **S. Wang**, S. Paul, J. Logan, and P. De Boeck, "Joint modeling with network analysis and psychometrics", *Geometry and Topology meet Data Analysis and Machine Learning, May, 2019.*

63. **S. Wang**, S. Paul, J. Logan, and P. De Boeck, "Joint analysis of social and item response networks with latent space models", *Algorithms for Threat Detection (ATD), November, 2018.*

64. **S. Wang** and P. De Boeck, "Does low reliability lead to low replication rates?", *International Meeting of Psychometric Society, July, 2018.*

65. **S. Wang**, B. Wiernik and B. Morgan, "Gender measurement bias versus trait differences in vocational interest assessment.", *Society for Industrial and Organizational Psychology conference, April, 2017*

66. C. Nye, J. Prasad, J. Bradburn, F. Elizondo, W. Phan, R. Amrhein, S. Cho, J. Rounds, B. Wiernik, **S. Wang**, B. Morgan, C. M.Wee, E. Wetzel, D. Ispas, "New directions for vocational interest research in organizations.", *Society for Industrial and Organizational Psychology conference, April, 2017*

## Colloquiums & Symposiums

67. P. Powla, S. Stephens, T. Sweet and †**S. Wang**, "The relationship between tau levels in functionally connected cortical brain regions and a region's tau burden.", *2024 IADRC Fall Research Symposium, October 2024.*

68. **S. Wang**, "Relationship between Changes in biomarkers and self-perceived functioning .", *Indiana Alzheimer's Disease Research Center, 2024.*

69. **S. Wang** and J. Edgerton, "Disrupting terrorist recruitment: Network knockouts of Islamic State recruiters.", *OSU Psychological Studies Quantitative Colloquium, 2021.*

70. **S. Wang** and J. Edgerton, "Disrupting terrorist recruitment: Network knockouts of Islamic State recruiters.", *OSU Political Science Studies Methodological Colloquium, 2021.*

71. **S. Wang** and P. De Boeck, "Understand reliability with the reliability paradox.", *OSU Psychological Studies Quantitative Colloquium, 2020.*

72. **S. Wang**, S. Paul, J. Logan, and P. De Boeck, "A joint modeling framework for social network analysis and psychometrics", *OSU Psychological Studies Quantitative Colloquium, 2019.*

73. **S. Wang** and P. De Boeck, "The role of reliability in the replication crisis.", *OSU Psychological Studies Quantitative Colloquium, 2018.*

74. **S. Wang**, "Applications of differential item functioning analysis with item response theory.", *Undergraduate Research Symposium, April, 2016.*

# Mentoring

## Postdoc Mentoring

- Jaeyoung Lee, Biostatistics and Health Data Science, IU School of Medicine (2026–present)
- Krishnendu Chandra, Biostatistics and Health Data Science, IU School of Medicine (2025–2026)
  - *Current position:* Lecturer, Texas A&M University

## PhD Dissertation Advisor

- Luling Altan, PhD in Biostatistics, Indiana University School of Medicine (2026–present)
- Renhao Li, PhD in Biostatistics, Indiana University School of Medicine (2026–present)

## Dissertation Committee Member

- Margaret Hawkins, PhD in Biochemistry and Molecular Biology, Indiana University School of Medicine (2026–present)

## Research Assistants Supervised

- Ash Sharma, Research assistant, Biostatistics and Health Data Science, IU School of Medicine (2025–present)
- Luke Hawthorn, INGEN4DS fellow and research assistant, Biostatistics and Health Data Science, IU School of Medicine (2025–present)
- Luling Altan, Research assistant, Biostatistics and Health Data Science, IU School of Medicine (2025–2026)
  - *Current position:* PhD student in Biostatistics, Indiana University School of Medicine
- Theyanesh Jayaprakash, Research assistant, Biostatistics and Health Data Science, IU School of Medicine (2024–2026)
  - Submitted one first-author manuscript and published one first-author abstract
  - *Current position:* AI Engineer, Plutus
- Connor Cornelison, Research assistant, Biostatistics and Health Data Science, IU School of Medicine (2024–2026)
  - Submitted one second-author manuscript and published two first-author abstracts; received INGEN4DS fellowship
  - *Current position:* Software developer supporting high-performance computing, Indiana University

- Huairui Wang, Research assistant, Biostatistics and Health Data Science, IU School of Medicine (2025–2026)
  - One first-author manuscript and one co-author paper submitted
  - *Current position:* PhD student in Biostatistics, Michigan State University
- Yumeng Chen, Research assistant, Biostatistics and Health Data Science, IU School of Medicine (2025–2026)
  - Submitted one manuscript
  - *Current position:* PhD student in Biomedical Engineering, Case Western Reserve University
- Gertrude Osei, Research assistant, Biostatistics and Health Data Science, IU School of Medicine (2025–2026)
  - Submitted one first-author manuscript and published one first-author abstract
  - *Current position:* PhD student in Educational Measurement, James Madison University
- Plamena Powla, PhD student, Biostatistics and Health Data Science, IU School of Medicine (2024–2025)
  - Published one first-author abstract
- Anastasija Naumoski, DataWiz intern, Biostatistics and Health Data Science, IU School of Medicine (Summer 2025)
- Yunhe Liu, Research assistant, Yale School of Public Health (2023–2024)
  - Published one co-author manuscript
  - *Current position:* PhD student in Statistics, Texas A&M University
- Yiting Wang, Research assistant, Yale School of Public Health (2023–2025)
  - Published two second-author manuscripts
  - *Current position:* PhD student in Statistics, University of Virginia
- Xinzhi Zhang, Research assistant, Yale School of Public Health (2023–2025)
  - Published one first-author and one co-author manuscript
  - *Current position:* PhD student in Biostatistics, New York University

# Service

## University Service

- Member, Admission Committee, Department of Biostatistics and Health Data Science, Indiana University 2025
- Member, Equality and Inclusivity Committee, Ohio State University 2021
- Member, Quantitative Psychology Group, Ohio State University 2019
- Member, Discussion Panel, the Office of Diversity and Inclusion, Ohio State University 2019
- Member, Psychology Diversity Council, University of Minnesota 2016
- Member, International and LGBTA: A Panel Discussion, University of Minnesota 2015
- Member, Orientation committee, International Student and Scholar Services, University of Minnesota 2015

## Professional Service

- Organizing Committee Member, ASA Statistics in Imaging Virtual Working Group 2025-
- Meeting Organizer, Regenstrief/IU AI conference 2025
- Session Chair, The Statistical Methods in Imaging Conference 2024
- Session Organizer, The Statistical Methods in Imaging Conference 2024
- Session Organizer, The Statistical Methods in Imaging Conference 2023
- Session Chair, International Meeting of Psychometric Society 2021

- Meeting Organizer, Translational Data Analytics Institute Fall Forum 2021
- Meeting Volunteer, American Statistical Association, the Caucus for Women in Statistics 2019
- Meeting Volunteer, State Science Day 2019
- Session Chair, International Meeting of Psychometric Society 2018

## Editorial Service

- Brain Informatics
- Imaging Neuroscience
- npj Dementia
- Journal of Eating Disorders
- Psychometrika
- Advances in Methods and Practices in Psychological Science
- Brain Structure and Function
- Brain and Behavior
- Journal of the American Statistical Association
- Medical Image Analysis
- International Conference on Bioinformatics and Biomedicine (BIBM2022)
- Scientific Report (4)
- Network Science
- Brain Imaging and Behavior (2)
- Alzheimer's & Dementia
- Biostatistics (2)
- Biometrics
- Journal of Educational Measurement
- Developmental Science
- Big Data Mining

## Peer Review: Competitions and Challenges

- Reviewer, Eastern North American Region Spring Meeting student paper competition 2026
- Reviewer, ICSA student poster competition 2025
- Reviewer, Eastern North American Region Spring Meeting student paper competition 2025
- Reviewer, The Statistical Methods in Imaging Conference student paper competition 2024
- Reviewer, National Council on Measurement in Education 2020